

Mallows's C_p

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In statistics, **Mallows's C_p** ,^{[1][2]} named for Colin Lingwood Mallows, is used to assess the fit of a regression model that has been estimated using ordinary least squares. It is applied in the context of model selection, where a number of predictor variables are available for predicting some outcome, and the goal is to find the best model involving a subset of these predictors.

Mallows' C_p has been shown to be equivalent to Akaike Information Criterion.^[3]

Definition and properties

Mallows's C_p addresses the issue of overfitting, in which model selection statistics such as the residual sum of squares always get smaller as more variables are added to a model. Thus, if we aim to select the model giving the smallest residual sum of squares, the model including all variables would always be selected. The C_p statistic calculated on a sample of data estimates the mean squared prediction error (MSPE) as its population target

$$E \sum_j (\hat{Y}_j - E(Y_j | X_j))^2 / \sigma^2,$$

where $\hat{Y}_j$ is the fitted value from the regression model for the j th case, $E(Y_j | X_j)$ is the expected value for the j th case, and σ^2 is the error variance (assumed constant across the cases). The MSPE will not automatically get smaller as more variables are added. The optimum model under this criterion is a compromise influenced by the sample size, the effect sizes of the different predictors, and the degree of collinearity between them.

If P regressors are selected from a set of $K > P$, the C_p statistic for that particular set of regressors is defined as:

$$C_p = \frac{SSE_p}{S^2} - N + 2P,$$

where

- $SSE_p = \sum_{i=1}^N (Y_i - Y_{pi})^2$ is the error sum of squares for the model with P regressors,
- Y_{pi} is the predicted value of the i th observation of Y from the P regressors,
- S^2 is the residual mean square after regression on the complete set of K regressors and can be estimated by mean square error MSE ,
- and N is the sample size.

Practical use

The C_p statistic is often used as a stopping rule for various forms of stepwise regression. Mallows proposed the statistic as a criterion for selecting among many alternative subset regressions. Under a model not suffering from appreciable lack of fit (bias), C_p has expectation nearly equal to P ; otherwise the expectation is roughly P plus a

positive bias term. Nevertheless, even though it has expectation greater than or equal to P , there is nothing to prevent $C_p < P$ or even $C_p < 0$ in extreme cases. It is suggested that one should choose a subset that has C_p approaching P ,^[4] from above, for a list of subsets ordered by increasing P . In practice, the positive bias can be adjusted for by selecting a model from the ordered list of subsets, such that $C_p < 2P$.

Since the sample-based C_p statistic is an estimate of the MSPE, using C_p for model selection does not completely guard against overfitting. For instance, it is possible that the selected model will be one in which the sample C_p was a particularly severe underestimate of the MSPE.

Model selection statistics such as C_p are generally not used blindly, but rather information about the field of application, the intended use of the model, and any known biases in the data are taken into account in the process of model selection.

References

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